Assignment 4: Model-Based Reinforcement Learning

Andrew ID: YOUR ID

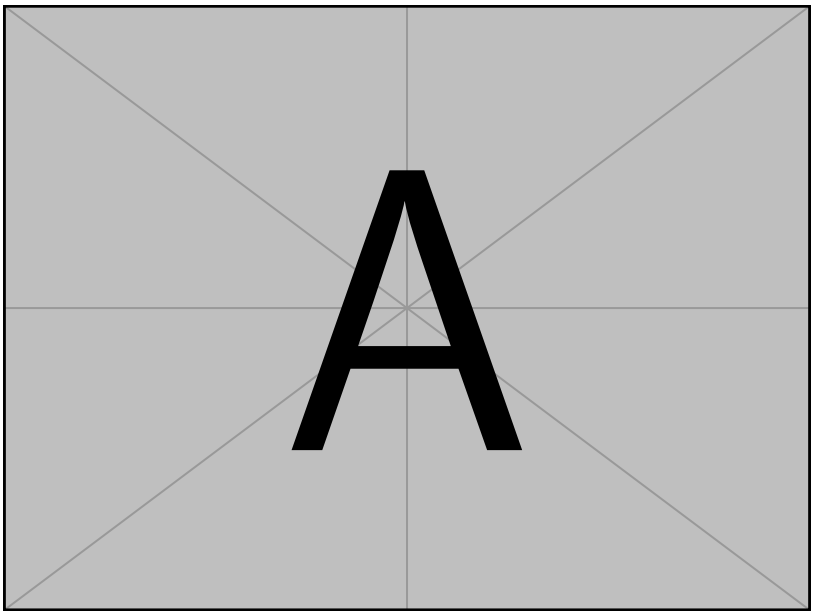
Collaborators (including AI tools, if applicable): YOUR COLLABORATORS

Total: 15 points. Keep the answer and plot sizes unchanged.

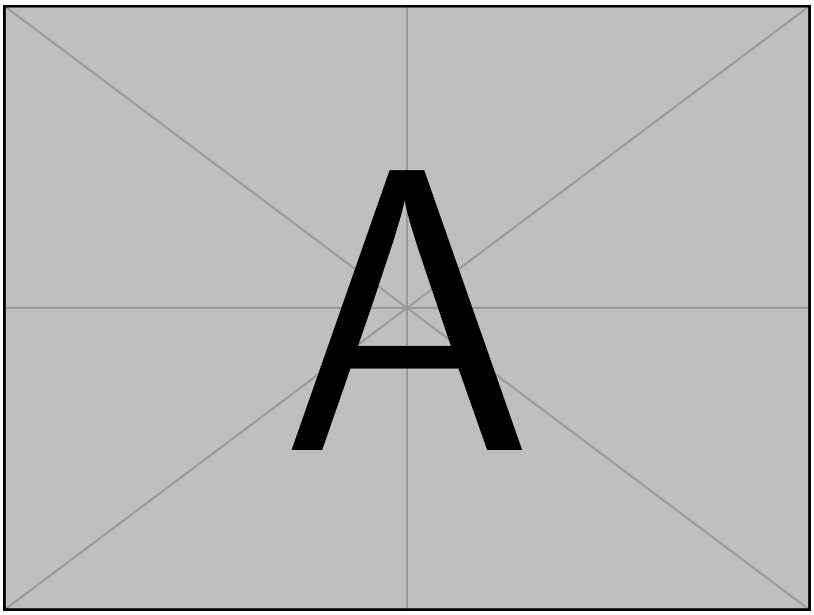
Problem 1: Dynamics Model Training (3.75 points)

Which model performs best, and why?

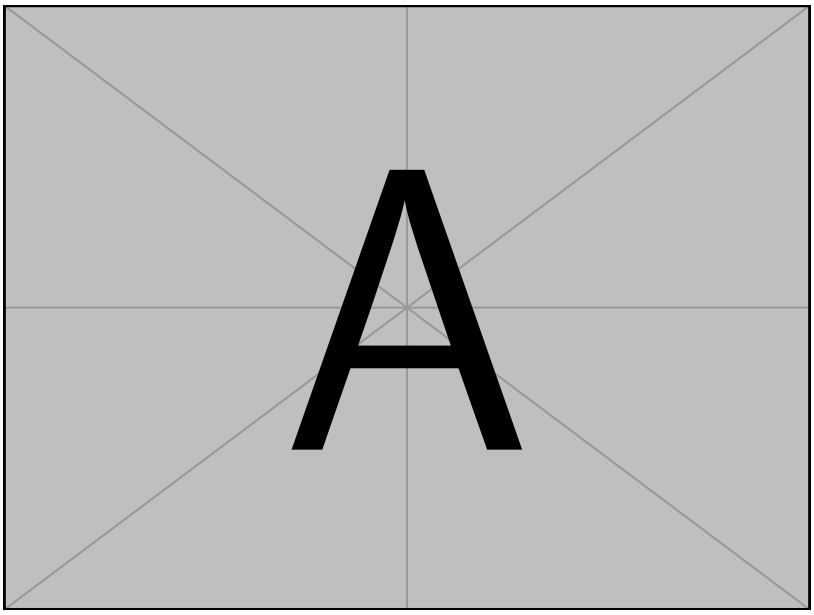
1 hidden layer, 32 units, 500 training steps



2 hidden layers, 250 units, 5 training steps

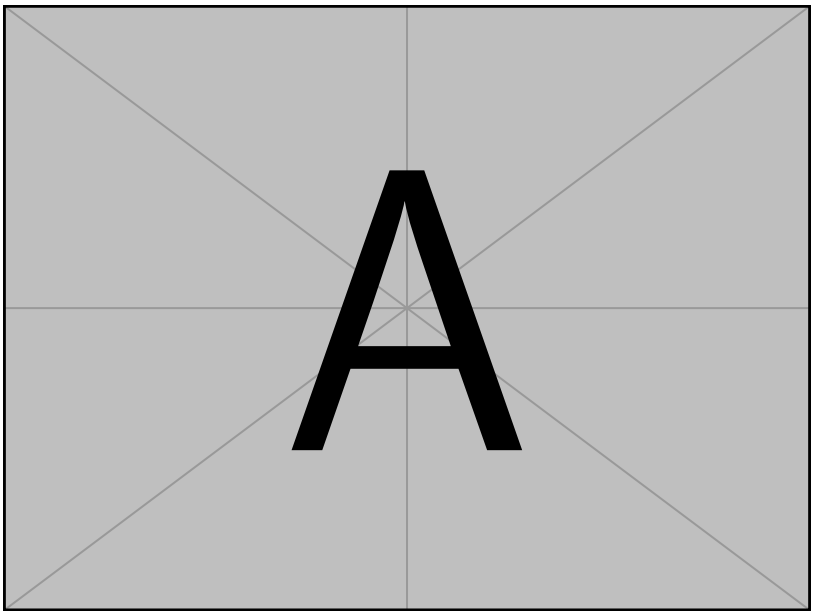


2 hidden layers, 250 units, 500 training steps



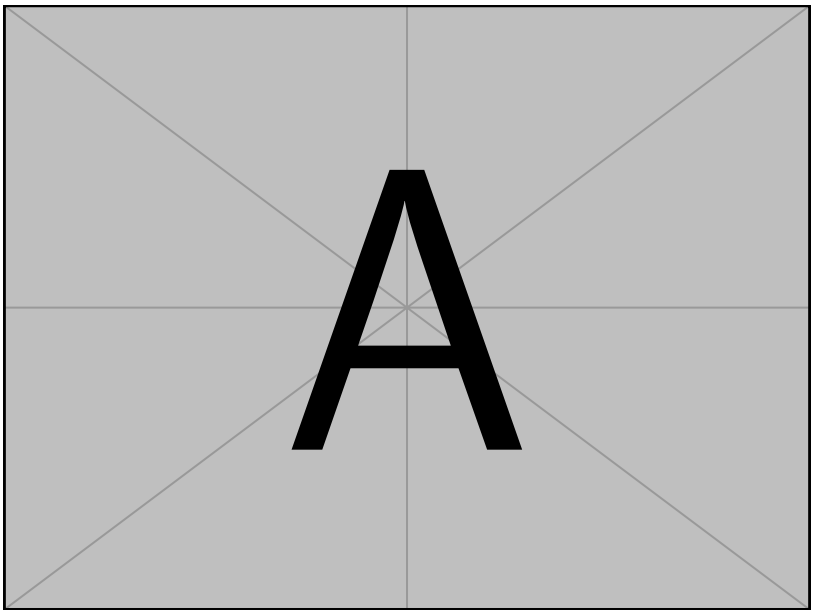
Problem 2: Action Selection (3.75 points)

Random actions versus MPC: training and evaluation returns

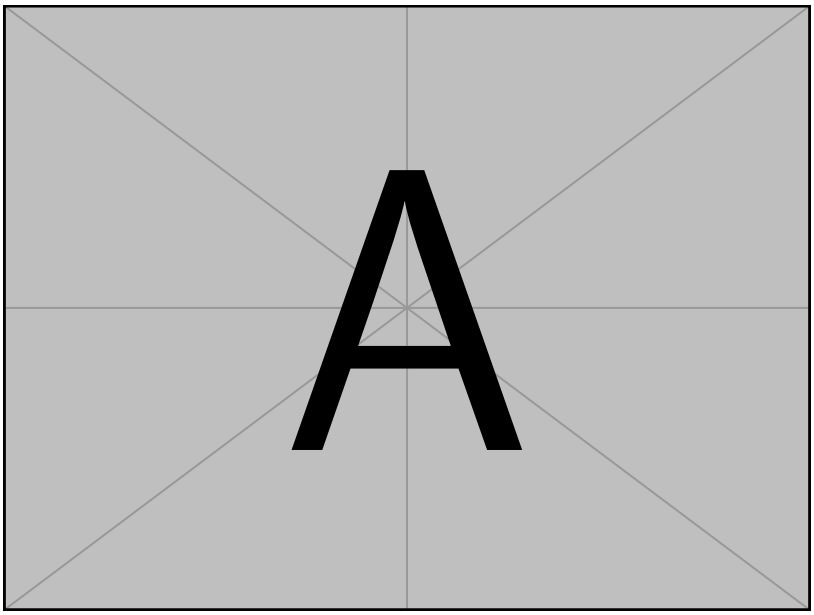


Problem 3: Iterative Model Training (3.75 points)

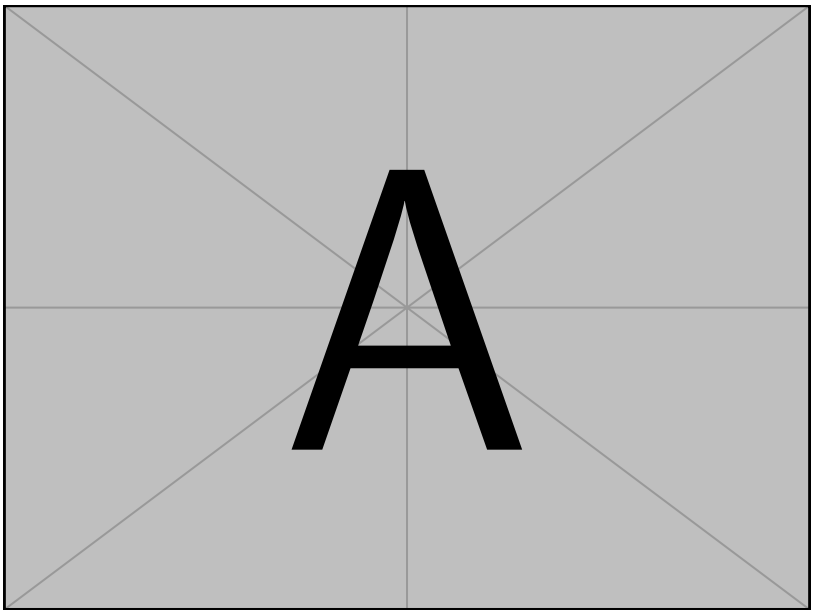
Obstacles learning curve



Reacher learning curve



Cheetah learning curve

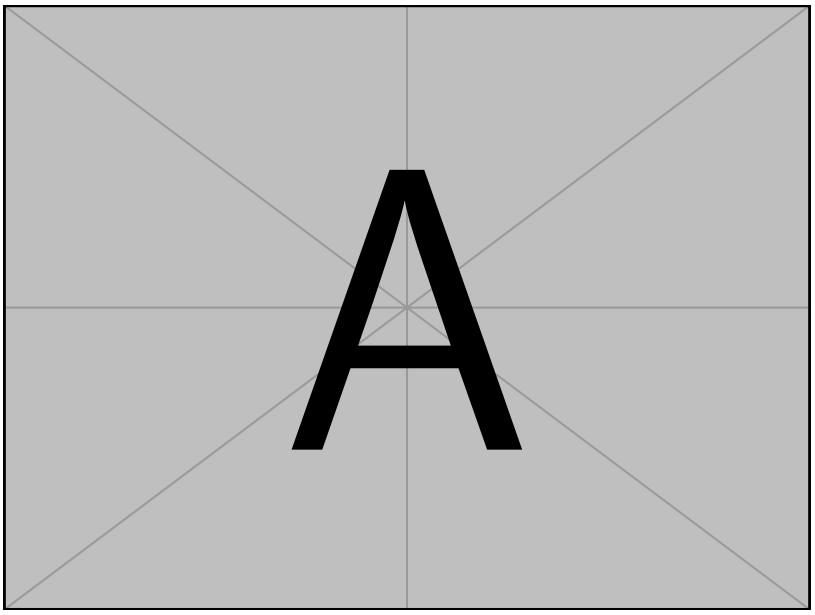


Problem 4: Hyperparameter Comparison and Discussion (3.75 points total)

Hyperparameter Comparison (3 points)

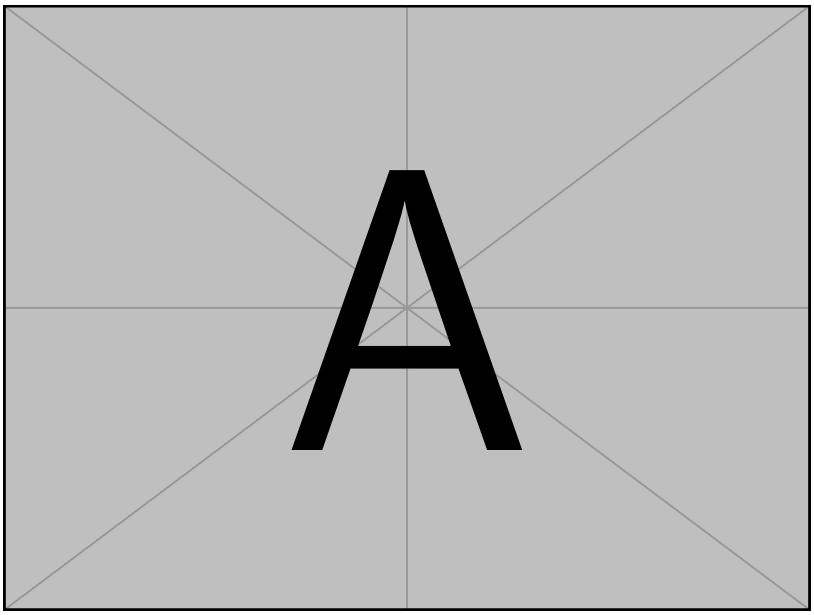
For each comparison, describe the observed trend and explain it in the description block directly below the plot.

Planning horizon: 5, 15, 30



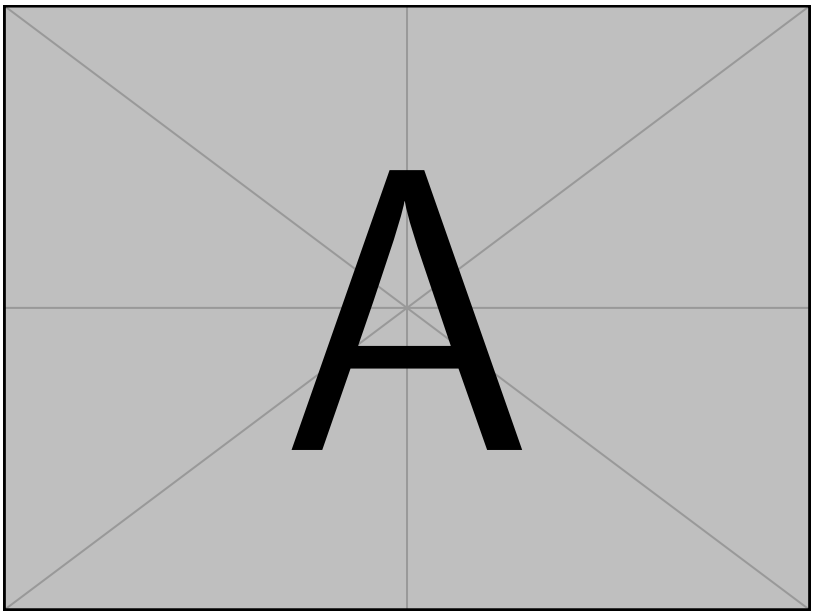
Description: planning horizon

Candidate sequences: 100, 1000



Description: number of candidate sequences

Ensemble size: 1, 3, 5



Description: ensemble size

Discussion (0.75 points)

Imagine you have a learned dynamics model that achieves low multi-step prediction error over the MPC planning horizon on held-out trajectories, but MPC using this model performs poorly. Give reasons why this could happen. Answer in 4–6 sentences.

Discussion answer